Maxwell J. Yin

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LinkedIn | GitHub | Portfolio

Summary

Ph.D. in NLP and Machine Learning Engineer specializing in LLM-powered product systems and evaluation. Experienced in designing and optimizing retrieval-augmented generation (RAG) systems, running large-scale experiments, and improving model quality, latency, and reliability in real-world applications.

Experience

Huawei Technologies - Noah's Ark Lab (Canada) 08/2025 – 02/2026

Machine Learning Engineer *Toronto, ON*

- **LLM Systems & Scaling:** Led 400M-parameter Transformer pretraining (40B tokens on 8×H100), delivering reproducible evaluation pipelines to support model selection and deployment decisions for downstream LLM applications (e.g., retrieval, reasoning, and user-facing AI systems) under compute constraints.
- **System Optimization & Deployment Readiness:** Profiled and resolved training/inference bottlenecks (GPU utilization, memory, latency). Optimized trade-offs to improve stability, throughput, and production readiness for downstream LLM applications.

The University of Western Ontario 09/2021 – 07/2025

Graduate Research Assistant *London, ON*

- **Production Semantic Retrieval:** Built scalable retrieval system over 500K+ documents / 1.7M passages, reducing latency by 5× while improving ranking quality, enabling faster and more reliable responses in downstream LLM applications.
- **Evaluation & Experimentation:** Designed and ran large-scale experiments to evaluate retrieval and generation quality, including ablation studies and benchmarking pipelines, identifying failure modes and enabling data-driven improvements in model robustness and performance.

Selected Projects

CineSeek: Production-Style Semantic Movie Search System 2026

Live Demo | GitHub

- **Retrieval-Augmented GenAI System:** Designed a production-style retrieval and ranking pipeline combining query rewriting, FAISS-based ANN search, and LLM reranking, optimizing latency-quality trade-offs for real-world semantic search applications.
- **Agentic Query Understanding:** Designed a production-style agent layer with HTTP-based tool invocation, replacing subprocess-based execution with a unified interface (MCP-style), enabling modular integration of retrieval and LLM components while reducing overhead and improving system reliability.
- **Full-Stack AI Deployment:** Built and deployed a production-style AI application with FastAPI backend, interactive React-based UI, and real-time query serving, containerized with Docker and deployed over HTTPS behind a reverse proxy, with CI/CD automation via GitHub Actions for reproducible deployment and continuous delivery.

Skills

- **LLM Product & Evaluation**: RAG systems, embedding retrieval, A/B testing, offline/online evaluation, failure analysis (hallucination, retrieval errors), prompt engineering, model quality benchmarking.
- **LLM Systems & Training**: Pretraining, inference optimization (KV cache, batching, vLLM), post-training, transformer architectures, DeepSpeed ZeRO, mixed precision.
- **Production Infrastructure**: Python, PyTorch, HuggingFace Transformers, FastAPI, model serving, scalable pipelines, FAISS, Docker, React, LangChain, OpenAI API, SLURM, AWS, Git.

Education

The University of Western Ontario 09/2021 – 07/2025

Ph.D. in Computer Science (Natural Language Processing) *London, ON*

- First-author publications in TACL, NAACL, and AAAI (Google Scholar).

The University of Western Ontario 09/2019 – 08/2021

Master of Science in Computer Science (NLP & Machine Learning) *London, ON*